

Finite-dice profile fluctuations at every bounded endpoint scale

Complete all-scales sharpness theorem; independently reviewed

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Theorem 1. *For every fixed $L > 0$ there are positive constants c_L, C_L such that the following holds for all sufficiently large m . For every scale $\lambda \in [m^{-1}, L]$, there is a fully permutation-fair family of $m + 1$ finite uniform dice and a face of one die whose comparison probabilities $p_1, \dots, p_m$ satisfy*

$$c_L \lambda \leq s := \sum_{i=1}^m (1 - p_i) \leq C_L \lambda, \quad \max_i |m(1 - p_i) - s| \geq c_L \sqrt{\lambda/m}.$$

All face labels can be chosen distinct. No bound on their number is asserted. The family may depend on both m and λ .

The theorem matches the square-root term in `natural_scale_endpoint_profile.tex` across all the displayed scales, up to fixed multiplicative constants in the realized s . It does not prescribe the exact numerical value $s = \lambda$. The separate example in `./endpoint_profile_sharpness.tex` has $s \asymp m^{-2}$ and error $\asymp m^{-1}$, forcing the additive floor as well.

Lemma 1 (A Gaussian node at the required scale). *Let $Q = \lceil (m + 1)/2 \rceil$ and let σ be the Q -node Gauss–Legendre probability rule on $[0, 1]$. For every $\lambda \in [m^{-1}, L]$ it has a node a of weight v with*

$$c\lambda/m \leq 1 - a \leq C\lambda/m, \quad v \geq c\sqrt{\lambda} m^{-3/2}.$$

The constants can be absolute once m is sufficiently large in terms of L .

Proof. Number the nodes downwards from 1, and choose $j = \lceil \sqrt{m\lambda} \rceil$. Since $m\lambda \geq 1$,

$$\sqrt{m\lambda} \leq j \leq 2\sqrt{m\lambda}.$$

For large m depending on L , $j \leq Q$ and $j/m \rightarrow 0$ uniformly over the stated scale range. Write $2a - 1 = \cos \theta$. The Legendre zero inequalities already used and justified in `./endpoint_profile_fluctuation_barrier.tex` are

$$\frac{(j - 1/2)\pi}{Q + 1/2} \leq \theta \leq \frac{j\pi}{Q + 1/2}.$$

They imply $\theta \asymp j/m$, uniformly even for $j = 1$. Thus $1 - a = (1 - \cos \theta)/2 \asymp j^2/m^2 \asymp \lambda/m$.

For completeness, the Bernstein–Legendre bound from the same reviewed note yields

$$\sum_{r=0}^{Q-1} (2r + 1) P_r(2a - 1)^2 \leq CQ / \sin \theta.$$

The Gaussian probability weight is the reciprocal of this diagonal reproducing kernel. Hence

$$v \geq c \sin \theta / Q \geq cj/m^2 \geq c\sqrt{\lambda} m^{-3/2}.$$

The two classical inequalities are recorded respectively in DLMF 18.16.2 and DLMF 18.14.7; the cited research note supplies the reproducing-kernel argument. $\square$

Proof of Theorem 1. The Gaussian rule integrates every polynomial through degree m , since $2Q - 1 \geq m$. Let F be its CDF. Its two one-sided values at the selected node a differ by v , so at least one, denoted F_* , satisfies $|F_* - a| \geq v/2$. Choose $t \in (0, 1)$ strictly between Gaussian nodes and sufficiently close to a on that side that

$$F(t) = F_*, \quad |t - a| < v/8, \quad |t - a| < (1 - a)/10.$$

Arbitrarily close one-sided approach makes all these requirements compatible. Consequently

$$|F(t) - t| \geq 3v/8, \quad c\lambda/m \leq 1 - t \leq C\lambda/m.$$

Consider the ideal row $(F(t), t, \dots, t)$ and write

$$s_0 = (m - 1)(1 - t) + (1 - F(t)).$$

Its lower bound is $s_0 \geq c\lambda$. For the upper bound it is enough to control the Gaussian tail; no cumulative quadrature-weight asymptotic is needed. Exactness of the m th moment gives

$$1 - F(t) = \sigma((t, 1]) \leq t^{-m} \int_0^1 x^m d\sigma(x) = \frac{t^{-m}}{m + 1} \leq \frac{C_L}{m}.$$

Indeed $1 - t \leq CL/m$, so $t^{-m} \leq \exp(C'L)$ for large m . Since $\lambda \geq 1/m$, this proves $s_0 \leq C_L\lambda$. The distinguished coordinate's ideal deviation is

$$|m(1 - F(t)) - s_0| = (m - 1)|t - F(t)| \geq cmv.$$

Apply the reviewed realization theorem `../finite_profile_realization.tex` to σ, t with

$$0 < \rho < \min\{v/100, c\lambda/(100m)\},$$

where $c > 0$ is small enough for the preceding lower bound on s_0 . Its hypotheses hold because t is not a Gaussian node. It produces an exactly permutation-fair family of finite uniform dice with a comparison row satisfying

$$|p_1 - F(t)| < \rho, \quad |p_i - t| < \rho \quad (i \geq 2).$$

Thus $|s - s_0| \leq m\rho$, preserving both comparability bounds. Writing $e_i = p_i - t$ for $i \geq 2$ gives

$$m(1 - p_1) - s = (m - 1)(t - p_1) + \sum_{i=2}^m e_i.$$

Its absolute value is at least

$$(m - 1)(3v/8 - 2\rho) \geq cmv \geq c\sqrt{\lambda/m},$$

as required. Distinct labels and uniform face probabilities are part of the realization theorem. $\square$

Theorem 2 (Sharp order at every bounded endpoint scale). *Define*

$$R_m(t) = \min\{mt, \sqrt{t/m} + m^{-1}\} \quad (t \geq 0).$$

For every fixed $L > 0$, the natural-scale upper theorem and the trivial bound give, for every admissible monotone tensor and $s \leq L$,

$$\max_i |m(1 - p_i) - s| \leq C_L R_m(s).$$

Conversely, for every sufficiently large m and every $\lambda \in (0, L]$, actual finite uniform permutation-fair dice have a comparison row with $c_L\lambda \leq s \leq C_L\lambda$ and

$$\max_i |m(1 - p_i) - s| \geq c_L R_m(\lambda).$$

As before, scale comparability is asserted, not the exact value $s = \lambda$, and there is no bound on the face counts.

Proof. Since $0 \leq u_i \leq s$, one has $|mu_i - s| \leq (m-1)s \leq ms$ for $m \geq 2$. Combine this with the natural-scale upper theorem to prove the upper bound. If $\lambda \geq 1/m$, Theorem 1 supplies the lower bound, since $R_m(\lambda) \leq 2\sqrt{\lambda/m}$.

For smaller scales use the Legendre construction from `../endpoint_profile_sharpness.tex`, with adjustable amplitude. Let $r \in \{m, m+1\}$ be odd and put $\varepsilon_0 = 1/[8r(r+1)]$. For $0 < \varepsilon \leq \varepsilon_0$, let σ_ε have mass ε at each endpoint and interior density

$$f_\varepsilon(t) = 1 - 2\varepsilon P'_r(2t-1).$$

The Legendre derivative bound gives $7/8 \leq f_\varepsilon \leq 9/8$, and the interior CDF is $F_\varepsilon(t) = t - \varepsilon P_r(2t-1)$. Integration by parts and Legendre orthogonality show that σ_ε has the uniform moments through degree m , exactly as in the cited note. Also

$$P_r(2t-1) \geq 1 - r(r+1)(1-t). \quad (1)$$

The ideal comparison row $(F_\varepsilon(t), t, \dots, t)$ has

$$s_0 = m(1-t) + \varepsilon P_r(2t-1), \quad |m(1 - F_\varepsilon(t)) - s_0| = (m-1)\varepsilon |P_r(2t-1)|.$$

If $0 < \lambda \leq \varepsilon_0$, choose $\varepsilon = \lambda$ and $t = 1 - \varepsilon/(100m)$. Equation (1) gives $P_r(2t-1) \geq 1/2$, so $s_0 \asymp \lambda$ and the deviation is at least $cm\lambda$. In this range $R_m(\lambda) = m\lambda$.

If $\varepsilon_0 < \lambda < 1/m$, choose $\varepsilon = \varepsilon_0$ and $t = 1 - \lambda/(100m)$. For all sufficiently large m , (1) again gives $P_r(2t-1) \geq 1/2$. Consequently

$$\lambda/100 \leq s_0 \leq \lambda/100 + \varepsilon_0 \leq 2\lambda, \quad |m(1 - F_\varepsilon(t)) - s_0| \geq c/m.$$

This is at least a fixed multiple of $R_m(\lambda)$, since $R_m(\lambda) \leq 2/m$ throughout the present range.

In both cases t is an interior continuity point of σ_ε . Apply the finite realization theorem with $\rho < \min\{\varepsilon/100, c\lambda/(100m)\}$, taking c smaller than the fixed lower comparison constant for s_0 . The error in s is at most $m\rho$, and the deviation of the marked coordinate changes by at most $2m\rho$. Thus all preceding lower and comparability bounds survive. The resulting dice are exactly fair, equiprobable, and have distinct labels. This proves the tiny-scale cases and completes the theorem. $\square$

Sharp synchronization threshold. The natural-scale upper bound gives uniform relative synchronization when $ms \rightarrow \infty$ on bounded endpoint ranges. This growth condition cannot be replaced by a bounded- ms assertion: choose $\lambda = 1/m$ above. Then $s \asymp 1/m$ and the absolute deviation is at least c/m , so $\max_i |m(1 - p_i)/s - 1|$ stays bounded away from zero.